

FairPlay DRM is lost during the process, so the MP3 files will play on any MP3 player. This means that anyone with even minimal computer skills can bypass the Apple iTunes implementation of DRM, admittedly with some possible loss in quality.

Apple adopted FairPlay in order to convince its content providers, the record companies that own the mechanical copyrights to the music recordings, that the revenue stream would be protected and that iTunes would not simply expose music tracks to wholesale copying. Comments made by Steve Jobs indicate that Apple would happily drop FairPlay, or any form of DRM, if the record companies would agree. So far Apple has had to put a fair amount of effort into protecting FairPlay just to keep its content providers, the record companies, happy. Other music download companies, such as RealNetworks, have made attempts to sell music files from their sites that will play on the iPod and other Apple devices, in effect files containing cracks of the FairPlay system. Apple has taken action against this, because if it doesn't it could upset its content providers and find itself open to prosecution by the Recording Industry Association of America (RIAA).

Despite all this, in April of 2007, EMI announced it would be dropping DRM for its online music sold through iTunes and Apple has quite recently started offering some files for download, albeit at a higher price, that have no DRM protection.

DVDs and CSS

The DRM system used for DVDs is called the Content Scrambling System (CSS). Introduced around 1996, it uses a relatively weak, proprietary 40-bit stream cipher algorithm. The weakness of the cipher is due, in part, to the US government regulations in place at the time the scheme was drafted, which prohibited the export of cryptographic systems using keys greater than 40 bits in length. A further structural weakness in the CSS encryption algorithm reduced the effective key length to only 16 bits, making it very easy to break. In fact this cipher was broken fairly early on and many programs are available for download that will strip out the encryption and produce a disk image for burning a copy.

All DVD players or drives include a CSS decryption module and the lead-in areas of all protected commercial DVD disks contain some encryption keys. These keys are stripped inside the drive during playback so a byte-for-byte copy made of the data stream output from a drive playing a commercial protected disk will not play back because the encryption keys aren't reproduced. All the CSS DRM is capable of doing is to provide weak protection against people copying commercial movies.



DRM & Audio CDs

In 2005, the Sony Bertelsmann Music Group (Sony BMG) included a content protection plan on its music CDs that was intended to stop consumers from using a computer fitted with a CD writer from copying Sony BMG music CDs. This caused something of a stink because, not only did the Sony BMG content protection involve planting rootkit code deep inside a PC's operating system, but it opened up a vulnerability that could be exploited by malware writers. Eventually, under pressure from various lawsuits, Sony BMG was forced to recall all the affected titles. In Europe in 2002, BMG had already caused an uproar by releasing copy protected titles that did not even carry a label stating that the CD incorporated copy protection.

Sony BMG employed two types of content protection software: Extended Copy Protection (XCP) and MediaMax CD-3. 52 titles were released with XCP and 50 titles were released with MediaMax. When a PC running a Windows operating system attempted to play one of these titles the Sony BMG software was triggered and would display a EULA message offering the user the option to install a custom Sony BMG player. Apparently the XCP software would install what came to be known as rootkit software, even before the EULA message was displayed.

Other criticisms of the Sony BMG software were that it ran as a constant background task, consuming system resources, it had no uninstall and could not be easily removed and that it caused instabilities leading to 'blue screen' crashes. Although Sony BMG did eventually release a supposed removal program, this software was ineffective and actually made things worse. These are the kind of problems that have led to all content protection and DRM getting a bad name.

In August 2000, in describing how serious Sony was about the copying issue, Sony Pictures Entertainment US senior VP Steve Heckler, told attendees at the Americas Conference on Information Systems. "The industry will take whatever steps it needs to protect itself and protect its revenue streams ... It will not lose that revenue stream, no matter what ... Sony is going to take aggressive steps to stop this. We will develop technology that transcends the individual user. We will firewall Napster at source — we will block it at your cable company. We will block it at your phone company. We will block it at your ISP. We will firewall it at your PC ... These strategies are being aggressively pursued because there is simply too much at stake."

Another, perhaps rather revealing comment, reportedly made by Peter Lee, an executive at Disney, was "If consumers even

know there's a DRM, what it is, and how it works, we've already failed,".

Sony BMG was not the only record label to release copy protected CDs. For example, Reprise Records, a Time Warner company, issued copy protected CDs.

Blu-ray, Advanced Access Content System (AACS) and HDCP

In theory and to the casual user, HDCP's combination of protected hardware links with encryption handshakes makes it impossible to copy. However the system can be attacked in two ways, either using a stripper box to extract the raw decoded signal, or by physically accessing the encoded data and decrypting it in software. A stripper box consists of some HDMI connectors wired to a circuit containing an HDCP decoder chip, something like the Silicon Image HD Fury chip. The problem with this method is that the particular box used can be remotely disabled by the key revocation tables issued on new media by the content providers. Decryption of the signal is not as difficult as it might have been because the economic burden of building really strong encryption into the hardware was felt by the hardware manufacturers to be too high.

Blu-ray disks and players use Advanced Access Content System (AACS) encryption; however as of August 2008 all current AACS decryption keys are available on the internet. BD+ and BD-ROM Mark are other encryption schemes that may be applied to Blu-ray and are tougher to crack than AACS.

Applications software

It's perhaps overlooked, but most commercial applications software has increasingly restrictive DRM based on a challenge and response scheme. Installation requires entering a serial number that ships with the product. This generates a further code which must be exchanged with the software vendor, either over the internet, or by phone, for a final unlock key. The installations are usually locked in to the PC the software is installed on and the maximum number of installs without further permission from the manufacturer is often limited. This system allows the software vendor to gather certain information about its customers.

Software End User Licence Agreements (EULAs) often state that you, as the end user, don't even own the physical software copy and documentation, but have only purchased a licence to use the software within the very narrow limits of the EULA, which the manufacturer and copyright owner may decide to revoke at any time. The final insult is that most software manufacturers include a clause in the EULA which says that they do not guarantee



their products to work and that they accept no responsibility for any negative consequences that may result from its use.

In conclusion

It's an accepted fact that rarity increases the worth of almost anything. If something is easily copied it is often regarded as being of low value. Also it seems that the ease with which something may be copied is directly related to the respect with which copyright is treated. Because digital assets, in particularly at the moment music tracks, are easy to copy, it has lead to music in general being devalued. Amazingly, many younger people seem to believe that music recordings should be free. What they do not seem to realise is that musicians, like everyone else, have to make a living. The eventual result of placing no value, or even a very low value, on music, or any other art form that can be recorded and copied, is that soon few will bother to make it. There will be no more, or at least very little, new music.

So despite their inconveniences, there is an argument for effective forms of DRM and the inconveniences of the current forms of DRM may well be short term. For example, at the moment PC owners, depending on the age of their machines, may need to update more components than just an optical drive to be able to play Blu-ray disks, purely because of High Definition Content Protection. As this technology finds its way onto the market and into new PCs this problem will go away. ■

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